

5.1 What are the main categories of materials available to designers when developing design solutions?

Understanding that products are predominantly made from multiple materials.

An overview of the **main categories** of materials as follows:

- a. papers and boards, including:
 - i. papers, e.g. layout and cartridge, different weights and coatings
 - ii. card, e.g. carton board, bleached card and corrugated card
 - iii. boards/sheets, e.g. foam board, Styrofoam and polypropylene sheet
 - iv. laminated layers, e.g. reflective surfaces.
- b. natural and manufactured timber, including:
 - i. hardwoods, e.g. oak, birch and teak
 - ii. softwood, e.g. pine, cedar and spruce
 - iii. manufactured boards, e.g. MDF, plywood and block board.
- c. ferrous and non-ferrous metals, including:
 - i. ferrous metals, e.g. iron, mild steel and stainless steel
 - ii. non-ferrous metals, e.g. aluminium, copper and tin
 - iii. alloys, e.g. brass, pewter and tin/lead solder.
- d. thermo and thermosetting polymers, including:
 - i. thermo polymers, e.g. PET, HDPE, PVC, LDPE, PS, PP, ABS, acrylic and TPE
 - ii. thermosetting polymers, e.g. silicone; epoxy resin and polyester resin.
- e. textile fibres and fabrics, including:
 - i. natural fibres, e.g. cotton, wool and silk
 - ii. synthetic fibres, e.g. nylon, polyester and acrylic
 - iii. mixed/blended fibres, e.g. cotton/polyester
 - iv. woven, non-woven and knitted fabrics.



- f. Awareness of developments in:
 - i. modern and smart materials such as graphene, super alloys, biopolymers and nano-materials
 - ii. composite materials and their purpose in relation to contrasting applications
 - iii. technical textiles used in different types of products dependent on context.

